from a few hundred to about 20000 base pairs. Polyacrylamide is preferred for the purification of smaller DNA fragments. The gel is complex network of polymeric molecules. DNA molecule is negatively charged molecule - under an electric field DNA molecule migrates through the gel. The electrophoresis is frequently performed with marker DNA fragments of known size which allow accurate size determination of an unknown DNA molecule by interpolation. The advantages of agarose gel electrophoresis are that the DNA bands can be readily detected at high sensitivity. The bands of DNA in the gel are stained with the dye **Ethidium Bromide** and DNA can be detected as visible fluorescence illuminated in UV light will give orange fluorescence, which can be photographed.

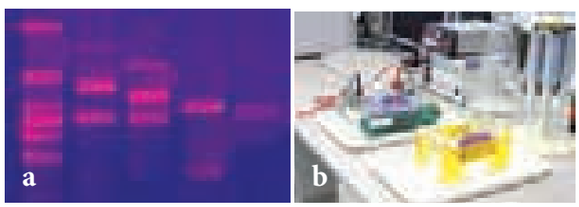

Figure 4.20: a. Bands of DNA in Agarose gel
b. Gel Electrophoresis Instrument

4.7.5 Nucleic Acid Hybridization - Blotting Techniques

Blotting techniques are widely used analytical tools for the specific identification of desired DNA or RNA fragments from larger number of molecules. Blotting refers to the process of immobilization of sample nucleic acids or solid support (nitrocellulose or nylon membranes.) The blotted nucleic acids are then used as target in the hybridization experiments for their specific detection.

Types of Blotting Techniques

Southern Blotting: The transfer of DNA from agarose gels to nitrocellulose membrane.

Northern Blotting: The transfer of RNA to nitrocellulose membrane.

Western Blotting: Electrophoretic transfer of Proteins to nitrocellulose membrane.

Agricultural diagnostics refers to a variety of tests that are used for detection of pathogens in plant tissues. Two of the most efficient methods are

1. ELISA (Enzyme Linked Immumo Sorbent Assay)

Elisa is a diagnostic tool for identification of pathogen species by using antibodies and diagnostic agents. Use of ELISA in plant pathology especially for weeding out virus infected plants from large scale planting is well known.

2. DNA Probes

DNA Probes, isotopic and non-isotopic (Northern and Southern blotting) are popular tools for identification of viruses and other pathogens

Southern Blotting Techniques - DNA

The transfer of denatured DNA from Agarose gel to Nitrocellulose Blotting or Filter Paper technique was introduced by Southern in 1975 and this technique is called Southern Blotting Technique.

Steps

The transfer of DNA from agarose gel to nitrocellulose filter paper is achieved by Capillary Action.

A buffer Sodium Saline Citrate (SSC) is used, in which DNA is highly soluble, it can be drawn up through the gel into the Nitrocellulose membrane.

By this process ss-DNA becomes '**Trapped**' in the membrane matrix.

This DNA is hybridized with a nucleic acid and can be detected by autoradiography.

Autoradiography - A technique that captures the image formed in a photographic emulsion due to emission of light or radioactivity from a labelled component placed together with unexposed film.

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